Potential Impact Assesment

Minor AI for Society

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1. Document History

Table 1. Document History

Version	Date	Status	Author	Description
0.1	09-03-2025	Initial Version	Casey Stijlaart	Initial version
0.2	12-03-2025	First Version	Casey Stijlaart	Added mitigation strategies and sustainability considerations
1.0	12-03-2026	Completed First Version	Casey Stijlaart	Restructured to reflect Responsible AI guidelines, future scenarios, and design reflections
2.0	12-06-2026	Revised Version, 2	Casey Stijlaart	Updated based on feedback, further reflections, and the bigger technological context

2. Purpose & Context

2.1. Impact on Society

The AI-powered Smart Desk Plant Monitoring system aims to reduce the “pain” of plant neglect and unsuccessful plant care, a common challenge for individuals in home and office environments By providing timely and personalized recommendations, the system addresses the problem of plant health deterioration due to inconsistent environmental awareness.

I am confident that the system addresses the right problem because the data collected (e.g., soil moisture, light, temperature, and humidity) directly informs actionable care recommendations. The system’s design focuses on user empowerment and education, rather than automation, to ensure that users remain engaged and informed about their plants’ needs. The intended societal outcome is improved environmental literacy, reduced plant loss, and increased engagement with small-scale environmental monitoring. I aim to avoid outcomes where users over-rely on AI, leading to reduced autonomy or misunderstanding of natural plant care processes.

Improvements considered: Implement adaptive recommendation messaging that emphasizes user judgment, integrate reminders to check sensor readings manually, and provide educational content about plant physiology.

2.2. Domain Understanding

Currently, plant care relies heavily on personal knowledge, habit, or generic online advice. Users vary widely in expertise, environmental constraints, and attention levels. Successful outcomes require consistent monitoring and correct interpretation of environmental signals; poor outcomes include plant death or chronic stress. Existing solutions (apps, watering reminders, environmental sensors) often fail due to over-reliance on cloud services, lack of context-specific guidance, or poor usability.

Assumptions: Users are willing to interact with a local dashboard, interpret suggestions responsibly, and accept minimal hardware installation. These will be validated through usability testing, and iterative prototypes.

Professional standards and constraints: Safety in electrical components, proper data protection, ethical AI communication, and sustainability considerations. Major uncertainties include user adherence, sensor accuracy, and long-term maintenance.

Improvements: Design for modular sensor replacement, ensure offline operation for privacy, and include clear instructions for user oversight.

2.3. Stakeholders

Primary stakeholders: Individuals who own plants at home or workspaces, affected by improved plant care and educational value.

Secondary stakeholders: Hobby gardeners, educators, and tech enthusiasts, affected by exposure to embedded AI applications and environmental awareness.

Indirect stakeholders: Environmental organizations, institutions exploring indoor environmental monitoring, or bystanders benefiting from improved ecological literacy.

Planned consultation: Prototype testing, and feedback loops to incorporate stakeholder insights.

Improvements: Broader engagement with diverse user groups to ensure recommendations are clear and actionable for novices and experts alike.

2.4. Bigger Technological Context

Looking at the bigger technological landscape, the Smart Desk Plant Monitoring system is part of a larger trend toward embedding Artificial Intelligence into physical environments. There is a growing push to integrate AI/ML directly into everyday objects and spaces, allowing them to sense, interpret, and respond to their surroundings in real time. This shift shows that AI is no longer confined to traditional computing platforms but is becoming a more seamless and integrated part of our physical world.

Embedded AI systems are increasingly being developed to improve the gap between humans and their environments, enabling more intuitive and responsive interactions. It has become more common for physical objects to be equipped with sensors and AI models that can interpret environmental conditions and communicate insights to users in an interpretable and interactive way.

Combining such aspects, sensors and machine learning models enable devices to translate environmental conditions into understandable feedback for users. In this setup, AI often acts like a bridge between people and their environment. Instead of users needing to interpret dashboards and data, the system can provide insights in a more natural and context-aware way, such as through subtle changes in lighting or notifications that appear when relevant. This approach can make technology feel more integrated into daily life, rather than something separate that requires constant attention.

This project showcases aspects of Ambient Intelligence and Ubiquitous Computing. Ambient Intelligence refers to digital systems that just run in the background of our daily lives, smartly reacting to what's happening around them and what we need. Ubiquitous Computing, on the other hand, describes a future where computing is built right into everyday objects and places, instead of being limited to just computers or screens. This Smart Desk Plant Monitoring system fits perfectly with these ideas because it puts sensing and AI directly into the physical space around the person using it.

On top of all that, the project also explores how we might slowly move away from constantly looking at screens to interact with technology. It suggests we could use more subtle, situation-aware ways of communicating. Instead of needing to check a dashboard all the time, this system can tell us about environmental conditions using simple, background cues like changes in lighting or notifications that appear just when they're relevant. This shows a bigger trend in embedded AI and how people use computers – moving towards technology that feels calmer and less pushy.

The basic ideas we used in this project can also be applied to all sorts of bigger areas in society and industry.

Examples include:

- Automotive systems capable of monitoring vehicle health, driver behavior, or environmental safety conditions.
- Smart homes that adjust lighting, climate, and energy consumption based on contextual AI analysis.
- Healthcare systems that monitor patient well-being through wearable or ambient sensing technologies.
- Industrial environments where embedded AI assists with predictive maintenance and operational safety.
- Energy management systems that optimize electricity usage and sustainability through real-time environmental monitoring.

What all these developments show is that embedded AI isn't just about automating tasks. It's more and more about making people more aware, helping them make better decisions, and improving how they interact with the places they live and work.

3. Societal Impact

3.1. Privacy

The system processes environmental data locally, no personal identifiers are collected. In case of dashboard integration requiring external access, anonymization and encryption are enforced.

Potential risks: Indirect profiling if users' patterns are combined with other datasets.

Improvements: Introduce strict data retention policies, anonymization safeguards, and user control over any transmitted data.

3.2. Data

Sensor data is objective but limited in context. Missing context or calibration issues may bias recommendations.

Mitigations: Continuous calibration, validation of sensors, and iterative model updates. Insights remain sustainable via adaptive models and regular data review.

Trust and consent: Users are informed about data use, consent is explicit for any external communication.

Improvements: Regularly review and document model assumptions, provide transparency in algorithm limitations.

3.3. Transparency

The system communicates plant recommendations as suggestions, not authoritative commands. Limitations and operational principles are clearly documented in the dashboard.

Improvements: Add FAQs, visual explanations of AI reasoning, and feedback channels for users to challenge outputs.

3.4. Inclusivity

Access may be limited by cost, technical literacy, or language barriers. Bias is minimal, as the system is largely universal for plant types, but cultural context may influence interpretation.

Improvements: Multi-language support, accessible interface design, and open-source options for wider adoption.

3.5. Sustainability

Local TinyML operation reduces cloud energy consumption, but materials like sensors and microcontrollers have environmental cost.

Improvements: Select durable, low-power components, design for repairability and modularity, and encourage recycling.

3.6. Hateful and Criminal Actors

The technology is low-risk for misuse, potential abuse includes misrepresentation of sensor data in shared environments.

Mitigations: Security of local devices, firmware validation, and clear disclaimers.

3.7. Human Values

Strengthened values include environmental awareness, autonomy in decision-making, and learning about AI and plants.

Improvements: Emphasize educational framing, provide autonomy in decision-making, and avoid gamification that pressures compliance.

4. Future Scenarios

4.1. Utopian Future

When looking into the future, for example the next 20 to 50 years, embedded AI systems will be integrated into everyday environments in a way that helps people understand and interact with their surroundings without the need for constant screen interaction and complex dashboards, which for most are too overwhelming. Examples of such could be small-scale AI devices assist users in understanding environmental conditions, sustainability, and resource usage with the help of subtle cues and feedback mechanisms, such as changes in lighting or notifications that appear when relevant.

This is a future where technology is more ambient and context-aware, providing feedback and insights without relying on constant screen interaction. Having humans become more environmentally aware and connected to their surroundings through embedded AI could lead to a more sustainable and harmonious relationship with nature. When one understands what is happening in their environment and how their actions impact the ecosystem, they can make more informed decisions that benefit both themselves and the planet. People are more willing to take care of their plants and the environment when they understand the conditions and how to improve them, leading to healthier plants and greener spaces. This could contribute to increased environmental awareness and urban greenery, as people become more connected to their plants and the environment around them. AI acts as a supportive and educational tool rather than replacing human decision-making, empowering users to learn and make informed choices about their environment.

4.2. Dystopian Future

When looking into a dystopian future, a future where embedded AI has severely negatively impacted both the environment and human values, many concerning things could be concluded.

Small systems such as the Smart Desk Plant Monitoring system could become heavily commercialized and over-adopted in everyday life. Users have become overly dependent on AI systems, reducing critical engagement with their surroundings and weakening environmental knowledge and practical skills. A extreme lack of awareness of their surroundings could lead to a disconnect from nature and a lack of understanding of how their actions impact the environment, which indirectly contributes to environmental degradation and a loss of biodiversity.

Something that could have an even bigger impact is the continuous replacement and adaptation cycle of embedded AI systems. In the dystopian future, systems that are not designed with long-term use in mind. Instead of allowing for long-term use, repairability, upgradability, and sustainability, they are designed to be replaced frequently with newer versions that have minor improvements or changes. This has caused a cycle of constant replacements, where users continuously throw out the old and replace it with the new. Overtime leading to the growing throwaway culture and significantly increased electronic waste.

As society becomes unable to keep up with recycling old electronic devices, pollution and resource depletion will become even more severe environmental problems.

5. Impact Reflection

5.1. Transformative Perspectives

By most people AI is seen as a software-based tool. A tool that can help us learn things, come up with ideas, or even help with writing a letter to a friend. Most people are not aware of how much more AI can be than just a software-based tool. When starting this semester, I was one of those people. When starting off, I mainly used AI as a tool to help me with writing, brainstorming, and learning about new topics. While I already knew that AI could be used in embedded systems, I didn't fully grasp how much it could influence the design and functionality of such systems, and how it could impact human values and interactions with technology.

As an embedded software engineer student, I tend to get stuck in the things that I know, things I know that work. While knowing the vital role AI plays within the project, I didn't realize to what extend. Until I had different conversations with my peers and consultants. They made me aware in what other ways AI could be used, and how it could influence the design of the system, and how it could impact human values and interactions with technology. This made me realize that design decisions around how AI can provide feedback to people without actually needing to know what is going on.

For example, when it comes to my personal project, the plant monitoring system, raw values and data are near useless to most people. They have no idea what to do with them, and they don't know how to interpret them. Taking this aspect and AI's ability to process data and provide feedback, I realized that the system could be designed to provide feedback in a more intuitive way. A way that does not require users to have all the deep knowledge about a plant. They see a green light, they know the plant is doing well, they see a red light, they know the plant needs attention. Just like with a stop light, red means stop, green means go, and yellow means slow down. This is a very intuitive way to communicate information, without the need for a screen or complex dashboard

To put it in perspective, the project made me realize that embedded AI systems may increasingly operate through ambient and context-aware interactions integrated into everyday life, allowing for a bridge to be created between people, technology, and the environment.

5.2. In Practice

At the start of the semester, I approached AI as a technical tool used through machine learning models to process sensor data and classify plant conditions. While I was aware of the potential for AI to be integrated into embedded systems, I didn't fully grasp the extent to which it could influence the design and functionality of such systems, and how it could impact human values and interactions with technology. So, the initial goal of the project was to create a functional and accessible monitoring system that could help users better understand the conditions of their plants through local AI processing and environmental sensing.

Once the idea was finalized, the project started off with finding and validating suitable datasets for training the machine learning model. While there were some datasets available, they were often limited in scope or not perfectly aligned with the specific conditions and sensors used in the project. This led to the decision to create a custom dataset based upon an existing dataset with engineered features that were more relevant to the project's goals. Upon initial validation, the model showed promising results, even with minor adjustments to the dataset and features.

Unfortunately, as the project progressed, it was discovered that the model was not as suitable as expected due to its heavy reliance on a single feature. This led to a revision of the dataset and the creation of a multi-feature model that could provide more robust and accurate feedback about plant health. This process improved understanding of how strongly dataset structure, feature engineering, and model design influence AI behavior and output.

As the project was nearing its original goal, the design evolved further to include an additional prediction model. This was decided based upon further discussions and reflections with consultants and peers, which made it clear that providing feedback on the current state of the plant was useful, but being able to predict how the plant may perform in the future could provide even more value to users. Not only would this be helpful for users, but it could also help bridge the gap between people and their environment by creating a stronger connection between users and their surroundings. This was an aspect that I gradually learned more about throughout the semester. So it could be said that over time, the role of AI within the project evolved from simply generating information about plant health and predictions towards acting as a bridge between environmental data and human understanding.

Using AI within an embedded system is valuable on its own, but making that AI meaningful and understandable to people could take it to another level. Something that could demonstrate how AI and technology may help people become more connected to their environment and more aware of it, rather than increasingly disconnected from it.

Over time, it became clear that the design of the system evolved from a relatively simple monitoring system into a broader embedded AI concept capable of both monitoring and predictive feedback. This development was driven not only by technical development, but also by a growing understanding of how AI could be used in more meaningful and human-centered ways within embedded systems.

The project ultimately highlighted the importance of flexibility and adaptability during the development process. Many aspects of the final system emerged through experimentation, reflection, technical limitations, discussions, and unforeseen challenges encountered throughout the semester.